

COURSE DESCRIPTION			
Program	Mechatronics		
Course Title	Fundamentals of Computer Programming		
Course Code	2432	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course introduces the fundamental concepts of programming using C language. It enables students to understand program structure, develop logic, use control structures, arrays, strings, functions, pointers, and file handling. The course helps students to develop problem-solving skills through programming.

Course Objectives

1	To introduce the fundamentals of programming using C language.
2	To develop problem-solving ability and logical thinking in students.
3	To familiarize arrays, strings, functions, pointers, and file handling concepts.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic computer knowledge Logical reasoning	1008	Foundational IT Skills	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Understand the fundamentals of C programming concepts and control structures.	Understand
CO2	Apply arrays, strings, structure and union	Apply
CO3	Develop modular programs using functions, recursion, searching and sorting techniques to solve problems.	Apply
CO4	Use pointers and file handling techniques effectively.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	1	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Total	12	-	1	-	-	-	-	-	-	-	-
Correlation Level	3	-	1	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
							CIE	ESE	Total	CIE	ESE	Total	
L	T	P	S										
3	1	0	0	5.5	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	C Fundamentals	• Introduction to programming paradigms • Applications of C language • Structure of a C program • Character set, constants, identifiers, keywords • Data types, variables • Expressions and statements • Input and output functions • Control statements ◦ if, if-else, nested if ◦ switch ◦ while, do-while, for ◦ break, continue	11	3
Module II	Arrays & Strings	Static and Dynamic memory Arrays • Declaration & Initialization • One-dimensional arrays • Array operations- Insert and Delete • Two-dimensional arrays Strings • String operations: length, compare, concatenate, copy • String matching • Built-in functions of string Structure and Union	11	4
Module III	Searching and Sorting	Function Function prototype, definition, and call • Add, subtract and multiply two numbers • Factorial of a number Recursion • Fibonacci series using recursion • Binary search using recursion Searching techniques • Linear search • Binary search Sorting techniques • Selection sort • Insertion sort • Bubble sort	12	4
Module IV	Function, Pointers & File	Pointers • Pointer operators • Parameter passing ◦ Pass by value ◦ Pass by reference Files • File concepts • Sequential file handling • Random access file handling • ● File operations using C - Read, Write and Append	11	4

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Introduction to programming paradigm, Need of programming	Understand what programming is and why C is widely used	L	CO1	Understand	1
Applications of C language	Identify real-time areas where C is used	L	CO1	Remember	1
Structure of C Program, Header files, main(), blocks	Explain structure of a C program	L/T	CO1	Understand	1
Character set, Constants, Identifiers, Keywords	Recall syntax rules and naming conventions	L	CO1	Remember	1
Data types & variables	Select appropriate data types for problems	L/T	CO1	Understand	1
Expressions & Statements	Write basic executable statements	L/T	CO1	Apply	1

Input/Output – printf(), scanf()	Use formatted input and output	L/T	CO1	Apply	1
Decision control: if, if-else, nested if, switch	Implement decision making in programs	L/T	CO1	Apply	2
Looping: while, do-while, for, break, continue, nested loops	Develop programs using loops	L/T	CO1	Apply	2
Module II					
Static and Dynamic memory	Explain the concept of memory	L/T	CO2	Understand	1
Array introduction	Explain need for arrays	L	CO2	Apply	1
Declaration & initialization	Use arrays in programs	L/T	CO2	Apply	1
One-dimensional arrays	Access and manipulate data	L/T	CO2	Apply	2
Two-dimensional arrays (matrix)	Work with matrix-based problems	L/T	CO2	Apply	1
Strings concept	Explain string storage and termination	L	CO2	Apply	1
String operations: length, compare, concatenate, copy	Demonstrate string handling	L/T	CO2	Apply	1
String matching programs	Develop basic string applications	T	CO2	Apply	1
Structure and Union	Explain the concept and implementation	L	CO2	Apply	2
Module III					
Function concept, Built-in library functions	Explain modular programming	L	CO3	Understand	1
Function prototype, definition & call	Write user-defined functions	L/T	CO3	Apply	2
Recursion	Explain concept of recursive calling	T	CO3	Understand	1
Recursive programs (factorial, Fibonacci)	Develop recursive programs	L	CO3	Understand	2
Searching: Linear & Binary search	Implement searching algorithms	L/T	CO3	Apply	2
Sorting: Selection, Insertion, Bubble sort	Apply sorting techniques	L/T	CO3	Apply	3
Module IV					
Pointer basics, declaration	Understand concept of memory referencing	L	CO4	Apply	1
Parameter passing	Implement different parameter passing	L/T	CO4	Understand	1
Call by value & call by reference	Implement call by value & call by reference	L	CO4	Apply	2
File concept & types - Sequential file processing – read, write, append	Explain file mechanism. Perform sequential operations	L	CO4	Understand	1
File concept of Random file	Explain file mechanism. Perform random operations	L	CO4	Understand	1
Concept of Files	Apply file operations techniques in programs	L	CO4	Apply	1
File operations using C	Read, Write and Append	L	CO4	Apply	4

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Understand different programming paradigms. Identify the structure of a C program	5
2	Explain the applications of C language. Use data types, variables, expressions, and identify the structure	5
Module II		
3	Understand static and dynamic memory. Work with one-dimensional and two-dimensional array	5
4	Perform array operations like insertion and deletion. Apply string operations and built-in string functions	5

Module III		
5	Identify prototype, definition, and call in a given program.	5
6	Write a program using a user-defined function. Use built-in functions from standard libraries. Identify appropriate header files for built-in functions.	5
7	Implement string length, copy, and compare without using library functions. Compare two strings and display whether they are equal. Compare recursive and iterative solutions.	5
8	Demonstrate pointer initialization and dereferencing Modify variable value using pointer. Explain pointer arithmetic with array examples. Traverse an array using pointers.	5.5
9	Pass array to a function using pointers. Declare and use a pointer to pointer. Display value using single and double dereferencing. Explain multi-level pointer concept with example.	5.5
10	Create an array of pointers to strings. Display multiple strings using array of pointers. Compare array of pointers with 2D character array.	5
11	Assign address of a function to pointer. Observe unchanged values after function call.	5
Module IV		
12	Merge two files into a third file.	5
13	Modify an existing record using random access	5
14	Implement efficient file handling techniques.	5
Total Self-Learning hours		70

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Programming in ANSI C	E. Balagurusamy	Tata McGraw Hill
2	Let Us C	Yashavant Kanetkar	BPB Publications
3	The C Programming Language	Brian W. Kernighan & Dennis M. Ritchie	Prentice Hall

REFERENCE			
Sl. No.	Title	Author	Publication
1	Computer Programming in C	V. Rajaraman	Prentice Hall India
2	C: How to Program	Paul Deitel & Harvey Deitel	Pearson Education

WEB RESOURCES		
Sl. No.	URL	Modules Covered
	NIL	

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60

Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

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